

Algebra: Graphing Logarithmic Functions

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DIRECTIONS

For each function, state the domain, range, asymptote, or identify key points.

1 Vertical asymptote:

$$f(x) = \log_4 x$$

Answer: _____

2 State the range:

$$f(x) = \log_2 x + 2$$

Answer: _____

3 Vertical asymptote:

$$f(x) = \ln x$$

Answer: _____

4 Evaluate one point:

$$f(x) = \log_2 (x + 3)$$

Answer: _____

5 x-intercept:

$$f(x) = \log_2 x - 3$$

Answer: _____

6 State the range:

$$f(x) = \log_{10} x$$

Answer: _____

7 Evaluate one point:

$$f(x) = \log_3 x$$

Answer: _____

8 State the domain:

$$f(x) = -\log_2 x$$

Answer: _____

9 State the range:

$$f(x) = \log_2 x$$

Answer: _____

10 Vertical asymptote:

$$f(x) = \log_2 (x - 2)$$

Answer: _____



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Answer Key & Solutions

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TEACHER NOTES Vertical asymptote shifts with horizontal translation. Range is always all real numbers.

1 Vertical asymptote:

$$= x = 0$$

$$f(x) = \log_4 x$$

2 State the range:

$$= (-\infty, \infty)$$

$$f(x) = \log_2 x + 2$$

3 Vertical asymptote:

$$= x = 0$$

$$f(x) = \ln x$$

4 Evaluate one point:

$$= (-1, 1)$$

$$f(x) = \log_2 (x + 3)$$

5 x-intercept:

$$= (1, -3)$$

$$f(x) = \log_2 x - 3$$

6 State the range:

$$= (-\infty, \infty)$$

$$f(x) = \log_{10} x$$

7 Evaluate one point:

$$= (3, 1)$$

$$f(x) = \log_3 x$$

8 State the domain:

$$= x > 0$$

$$f(x) = -\log_2 x$$

9 State the range:

$$= (-\infty, \infty)$$

$$f(x) = \log_2 x$$

10 Vertical asymptote:

$$= x = 2$$

$$f(x) = \log_2 (x - 2)$$

